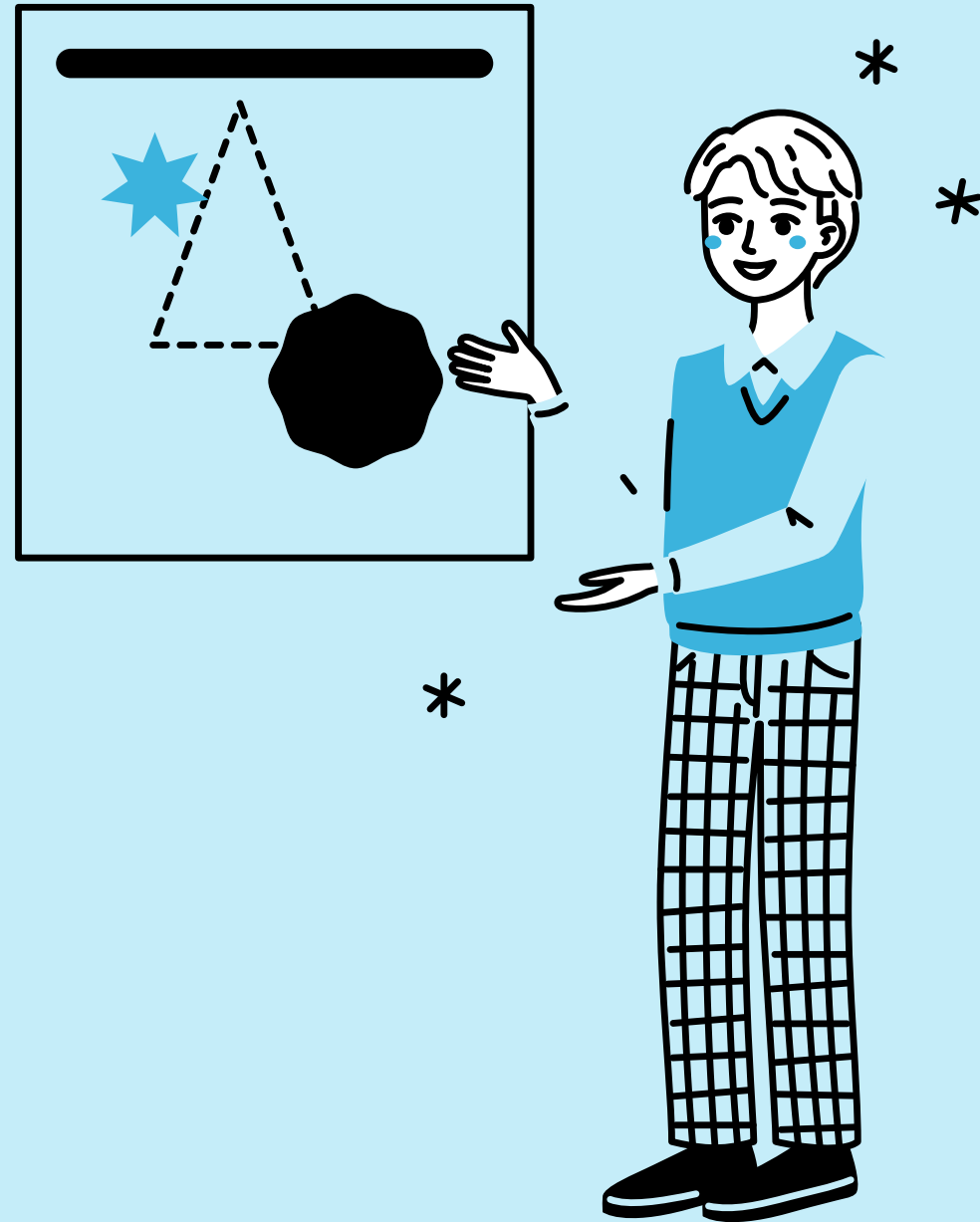


Employee Attrition Prediction –Machine Learning

Group Members:

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Obinna Ekere
Rilwan Ayoola
Emmanuella Adwubi Appiah

PROJECT OVERVIEW



Introduction



Data Cleaning & Exploration



Data Transformation



Exploratory Data Analysis (EDA)



Predictive Models



Model Selection

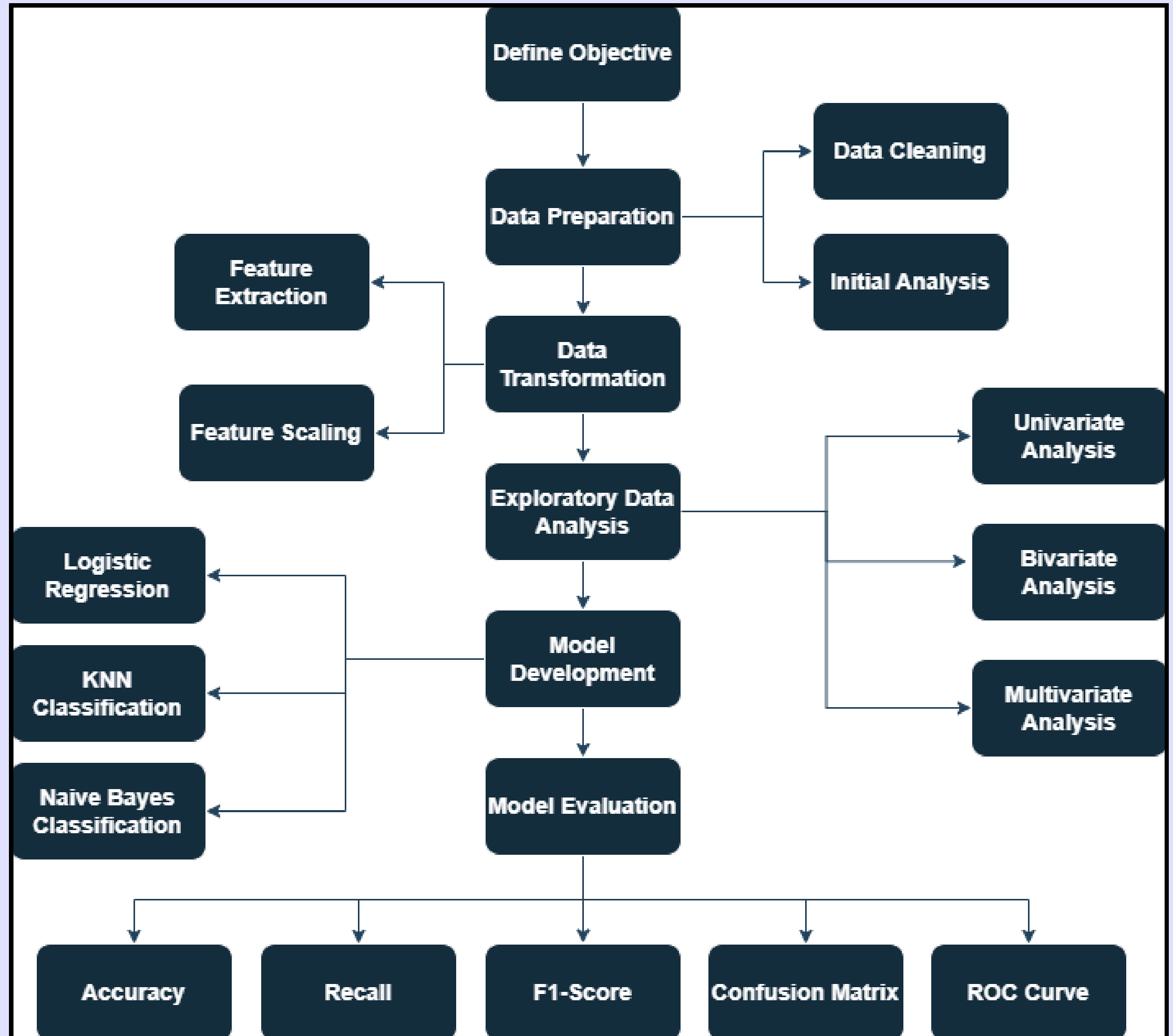


Conclusion

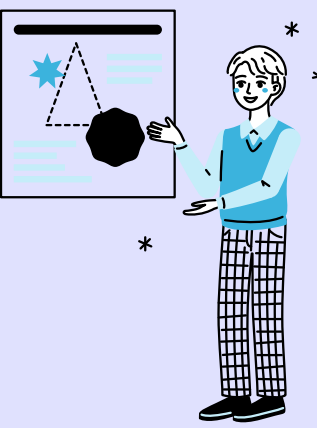


Methodology Flowchart

- Understand Objective
- Organize data
- Identify key drivers
- Predict attrition
- Measure Model



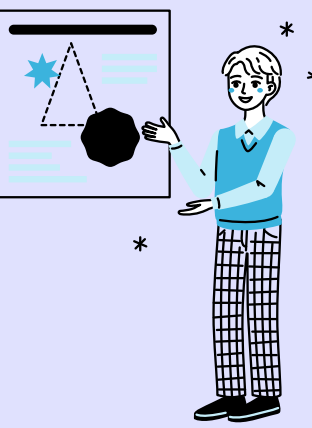
INTRODUCTION



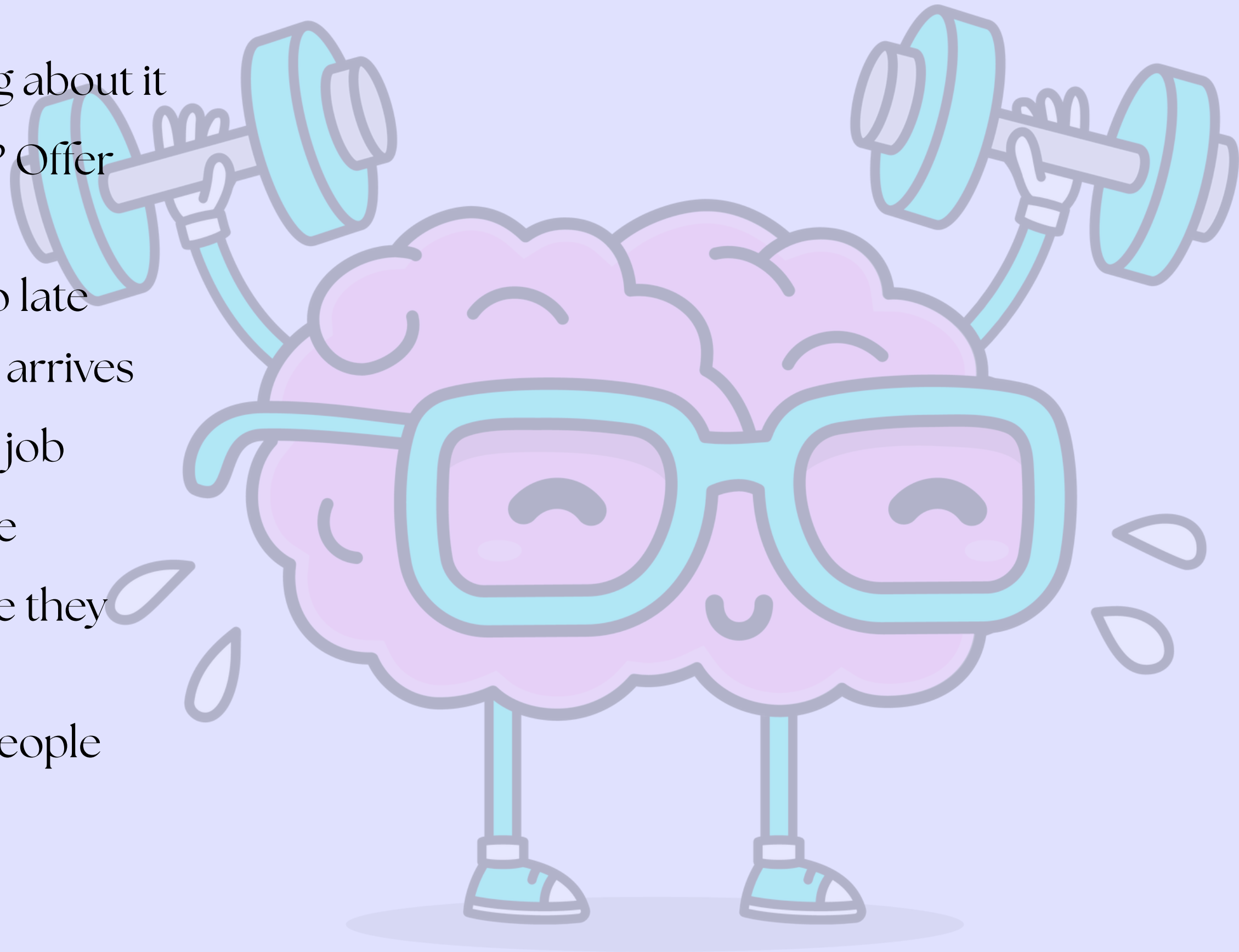
- Employee turnover = expensive problem for companies
- Hiring costs, training costs, lost productivity
- Team morale drops when good coworkers leave
- We used a dataset with 1,470 employees
- Data includes: age, salary, overtime, job satisfaction
- Goal: Build a model to predict who might quit
- Think of it as an early warning system for HR



PROJECT MOTIVATION



- If we know early → we can do something about it
- Give a raise? Improve work-life balance? Offer promotion?
- Right now, most companies find out too late
- They only know when resignation letter arrives
- By then, employee already has another job
- We want HR to be proactive, not reactive
- Machine learning = see problems before they happen
- Saves money, saves time, keeps good people



- No Duplicate Records
- No Missing Values
- Unique Records count
- Irrelevant columns dropped:
 - a. EmployeeNumber
 - b. EmployeeCount
 - c. Over18
 - d. StandardHours

- No Duplicate Records
- No Missing Values
- Unique Records count
- Irrelevant columns

dropped:

a. EmployeeNumber

b. EmployeeCount

c. Over18

d. StandardHours

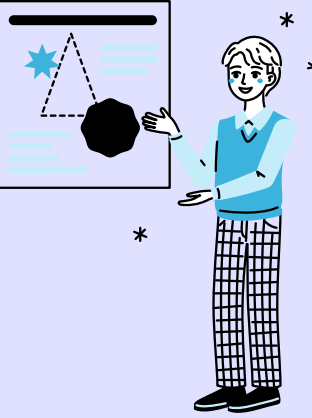
Duplicates: 0

Age	0
StandardHours	0
NumCompaniesWorked	0
Over18	0
OverTime	0
PercentSalaryHike	0
PerformanceRating	0
RelationshipSatisfaction	0
StockOptionLevel	0
MonthlyIncome	0
TotalWorkingYears	0
TrainingTimesLastYear	0
WorkLifeBalance	0
YearsAtCompany	0
YearsInCurrentRole	0
YearsSinceLastPromotion	0
MonthlyRate	0
MaritalStatus	0

EmployeeCount	1
EmployeeNumber	1470
EnvironmentSatisfaction	4
Gender	2
HourlyRate	71
JobInvolvement	4
JobLevel	5
JobRole	9
JobSatisfaction	4
MaritalStatus	3
MonthlyIncome	1349
MonthlyRate	1427
NumCompaniesWorked	10
Over18	1
OverTime	2
PercentSalaryHike	15
PerformanceRating	2
RelationshipSatisfaction	4
StandardHours	1

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction	Gender	...
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	2	Female	...
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	3	Male	...
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	4	Male	...
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	4	Female	...
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	Male	...
5 rows × 31 columns											

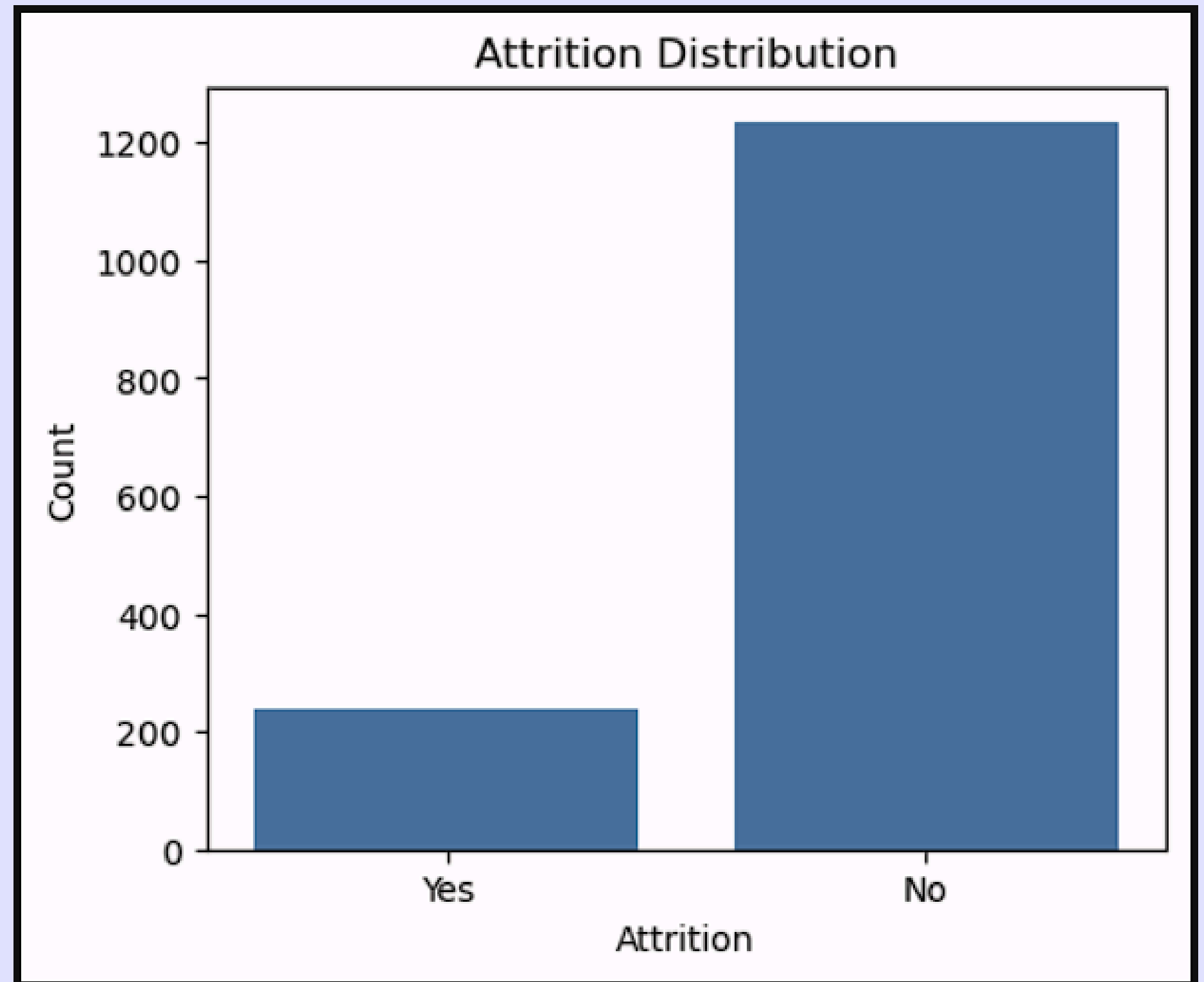
Initial Visualizations



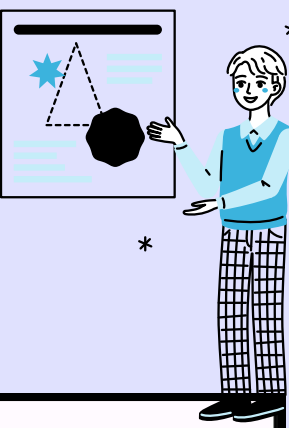
1. Attrition Distribution

- Target Variable
- Binary Outcome: Yes/No
- Low attrition rate

```
Attrition rate (proportion):  
Attrition  
No      0.838776  
Yes      0.161224  
Name: proportion, dtype: float64
```



2. Outliers Identification through boxplot

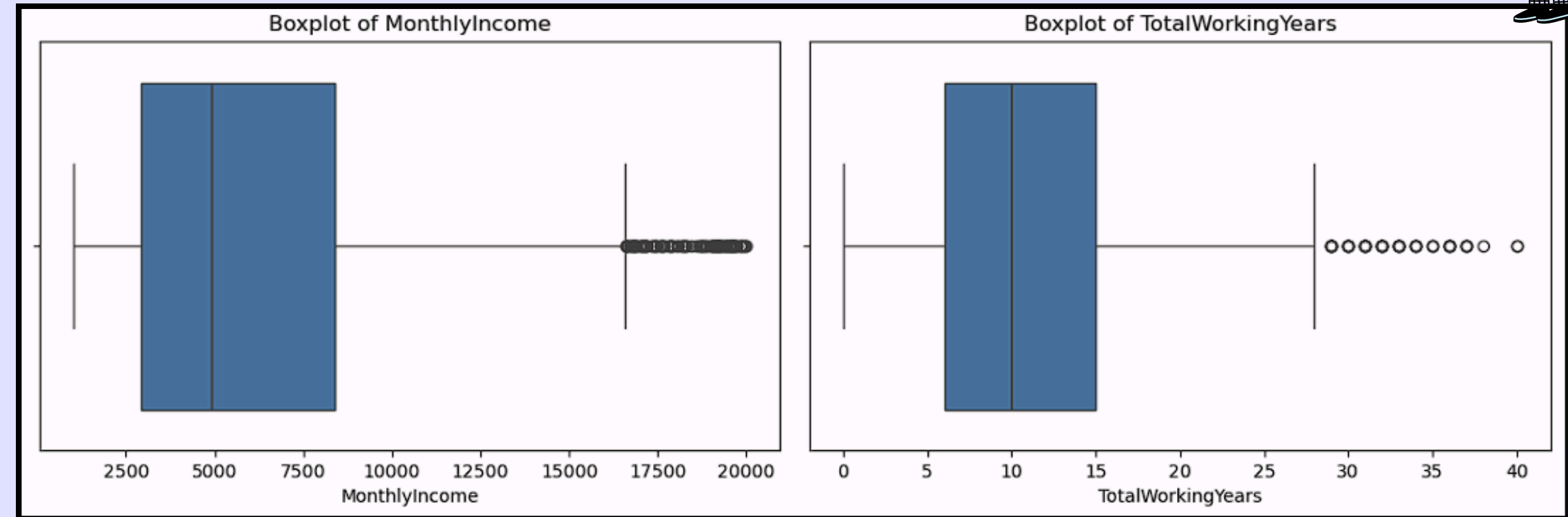


MonthlyIncome

- High Income outliers with wide spread
- Most employees earn in mid-range
- Some extreme high earners

TotalWorkingYears

- Presence of long-tenure outliers who have worked for 30+ years
- Most employees with experience less than 15 years
- Varying experience across workforce

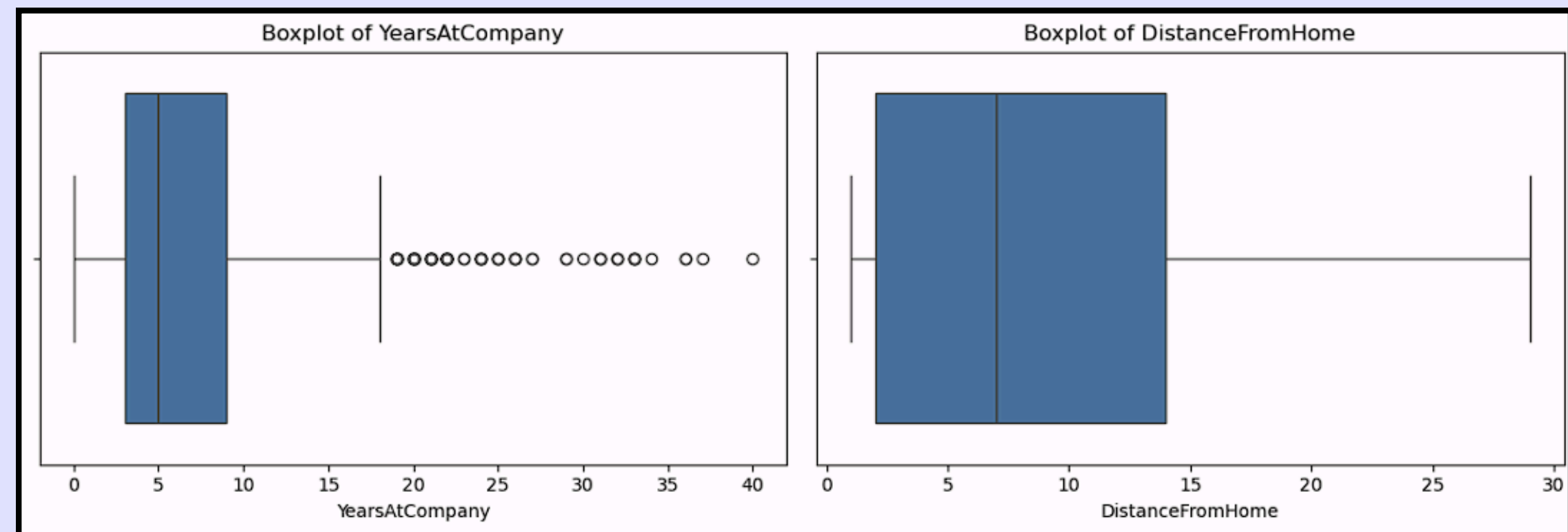


YearsAtCompany

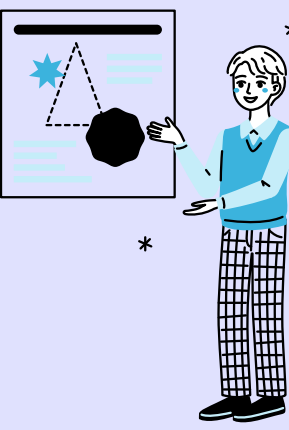
- Majority have short to mid tenure
- Long tenure outliers evident
- Employees with median spread of 5 years

DistanceFromHome

- Distribution moderately spread
- Majority lives close to workplace
- Few commutes from long distance



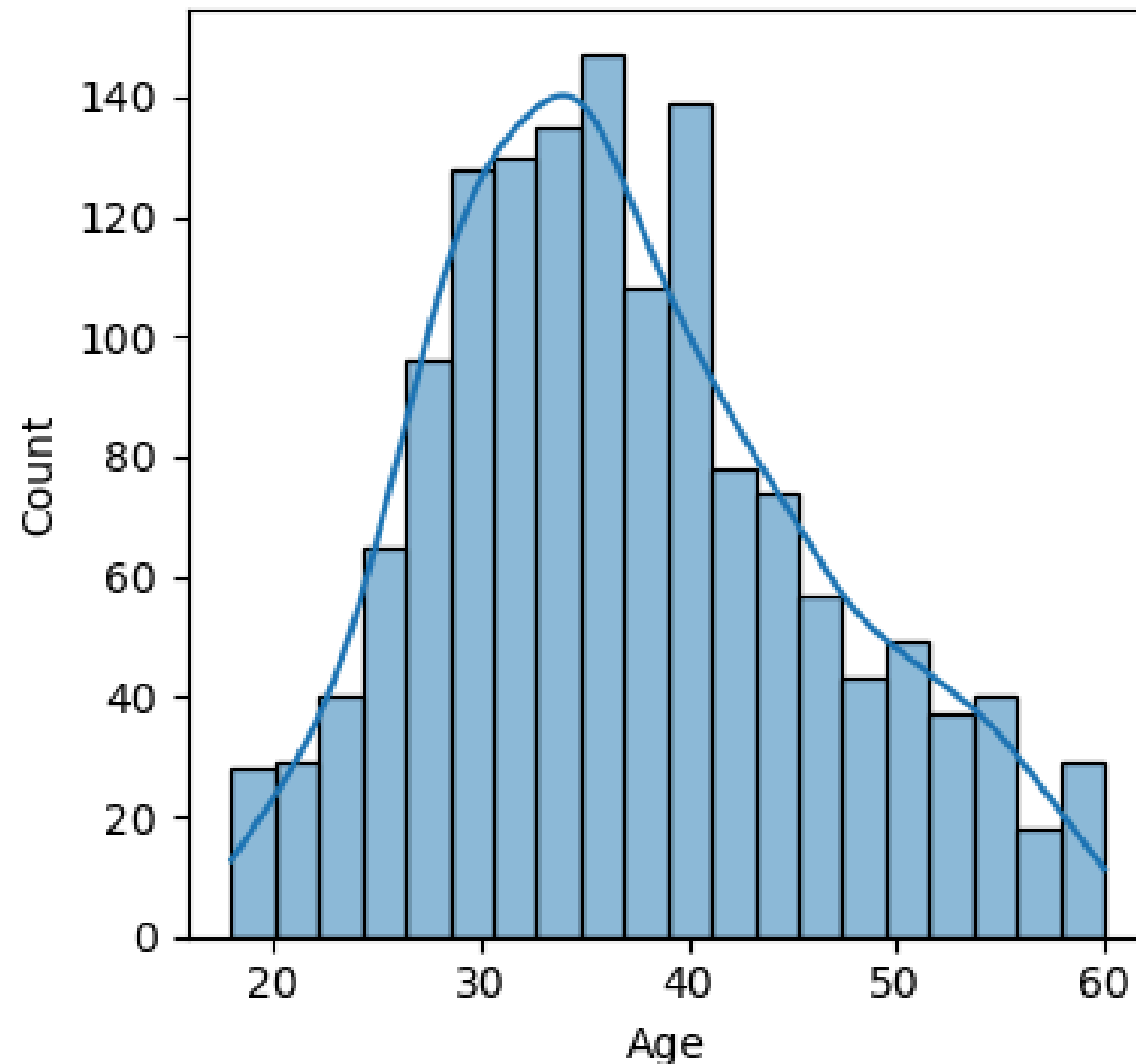
3. Skewness Check



Age

- Roughly symmetric
- Most employees at mid 30s
- Some approaching retirement age

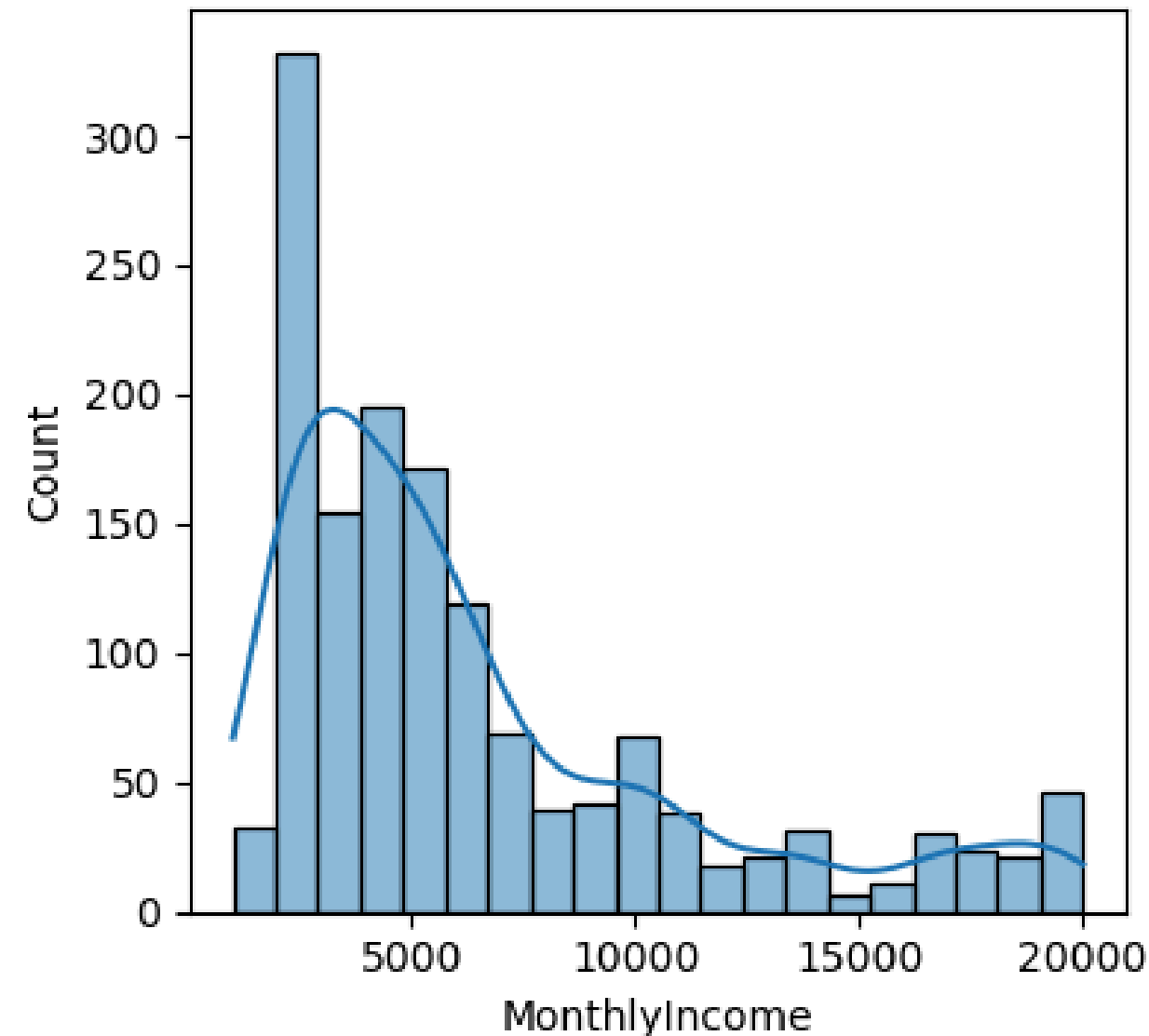
Distribution of Age



MonthlyIncome

- Right-skewed distribution
- Most employees in less than 5000 range
- Few high-income earners stretches the tail

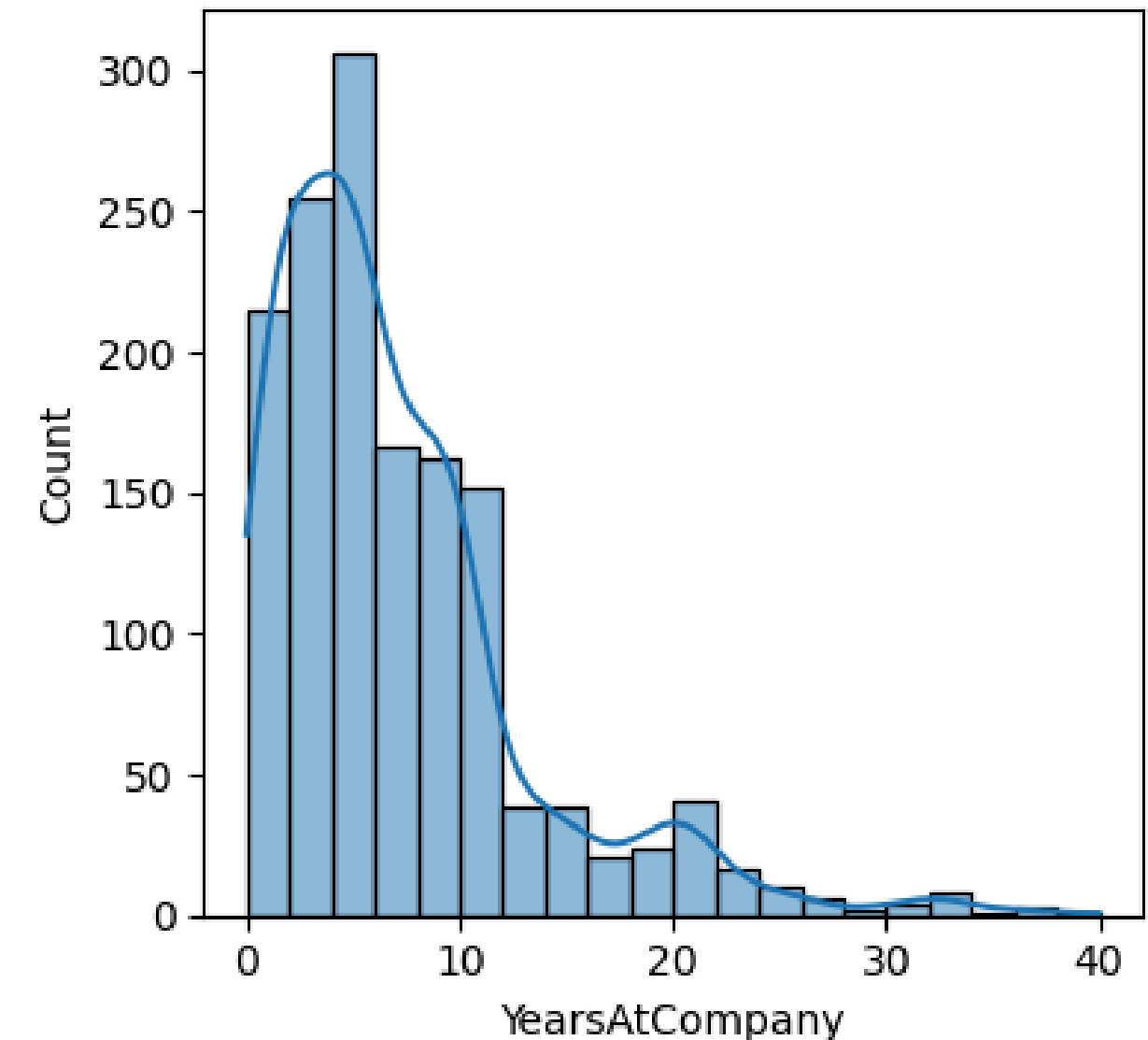
Distribution of MonthlyIncome



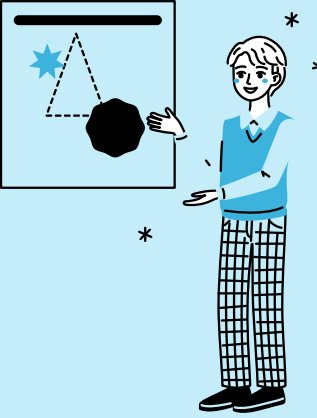
YearsAtCompany

- Right-skewed distribution
- Most employees have short tenure
- Few long-tenured employees forming tail

Distribution of YearsAtCompany



Data Transformation



Process

Feature Extraction



01

Binary and one-hot encoding:
Converting categorical to numerical

Feature Scaling



02

Normalize data using Standard Scaler.
Purpose for variables to contribute
equally

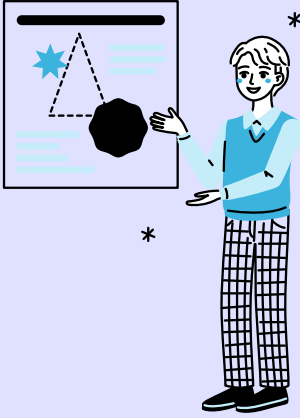
Train-test split



03

70/30 Split
Purpose: To evaluate model on unseen
data

Exploratory Data Analysis (EDA)



Univariate Analysis

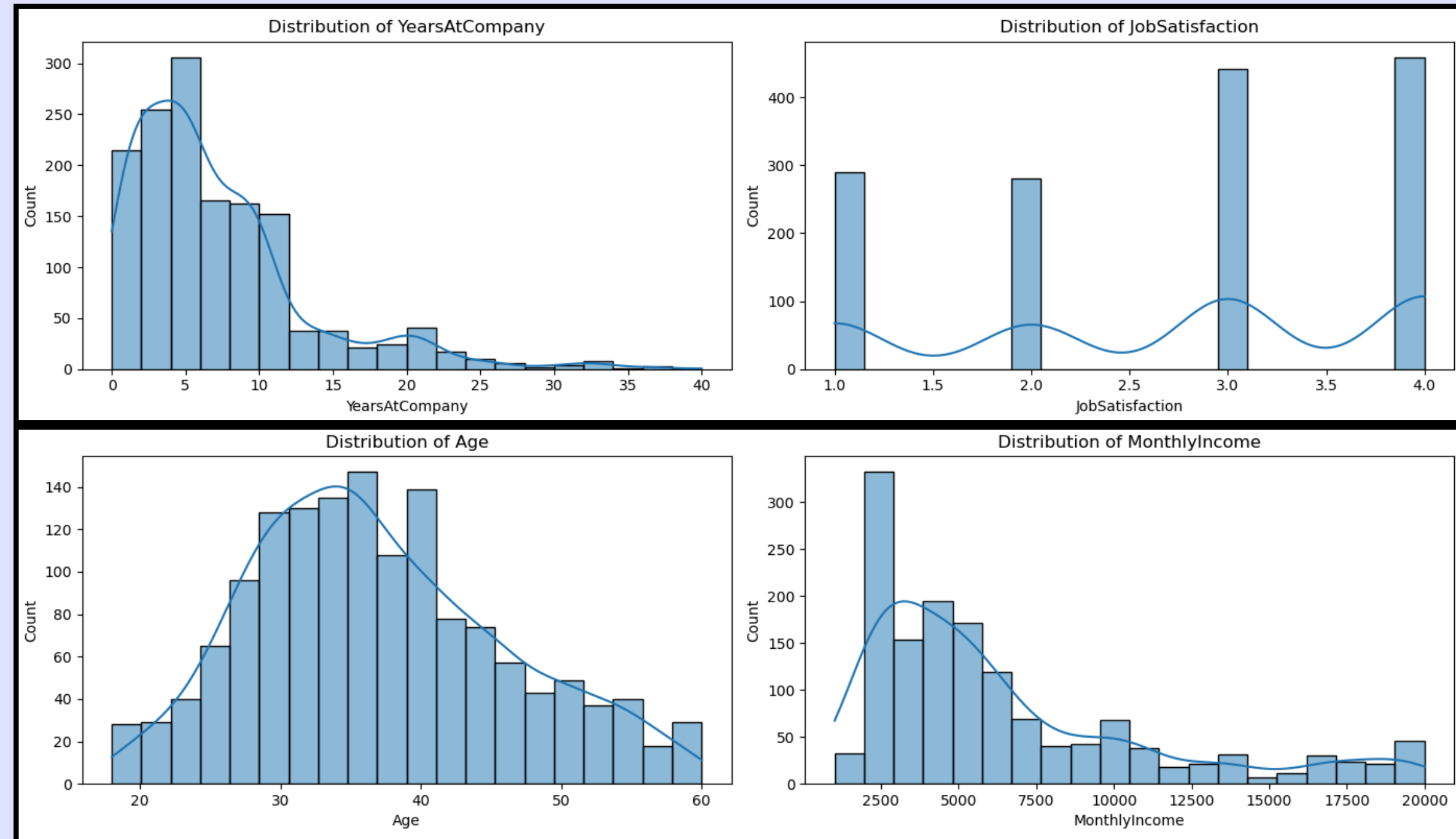
–YearsAtCompany, JobSatisfaction, Age, MonthlyIncome

Analysis:

- Years at Company is right-skewed, with most employees having short tenure
- Job Satisfaction shows peaks at certain levels, indicating uneven distribution
- Age distribution is approximately normal, centered around mid-age employees
- Monthly Income is highly right-skewed, with most employees earning lower salaries

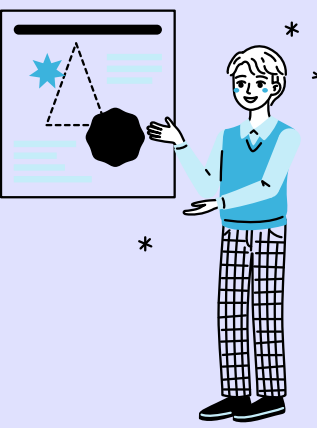
Insights:

1. These distributions suggest that employee attrition may be influenced by factors such as tenure, income level, and job satisfaction.
2. Understanding these individual patterns provides a foundation for identifying key drivers of attrition in later analysis.



Bivariate Analysis

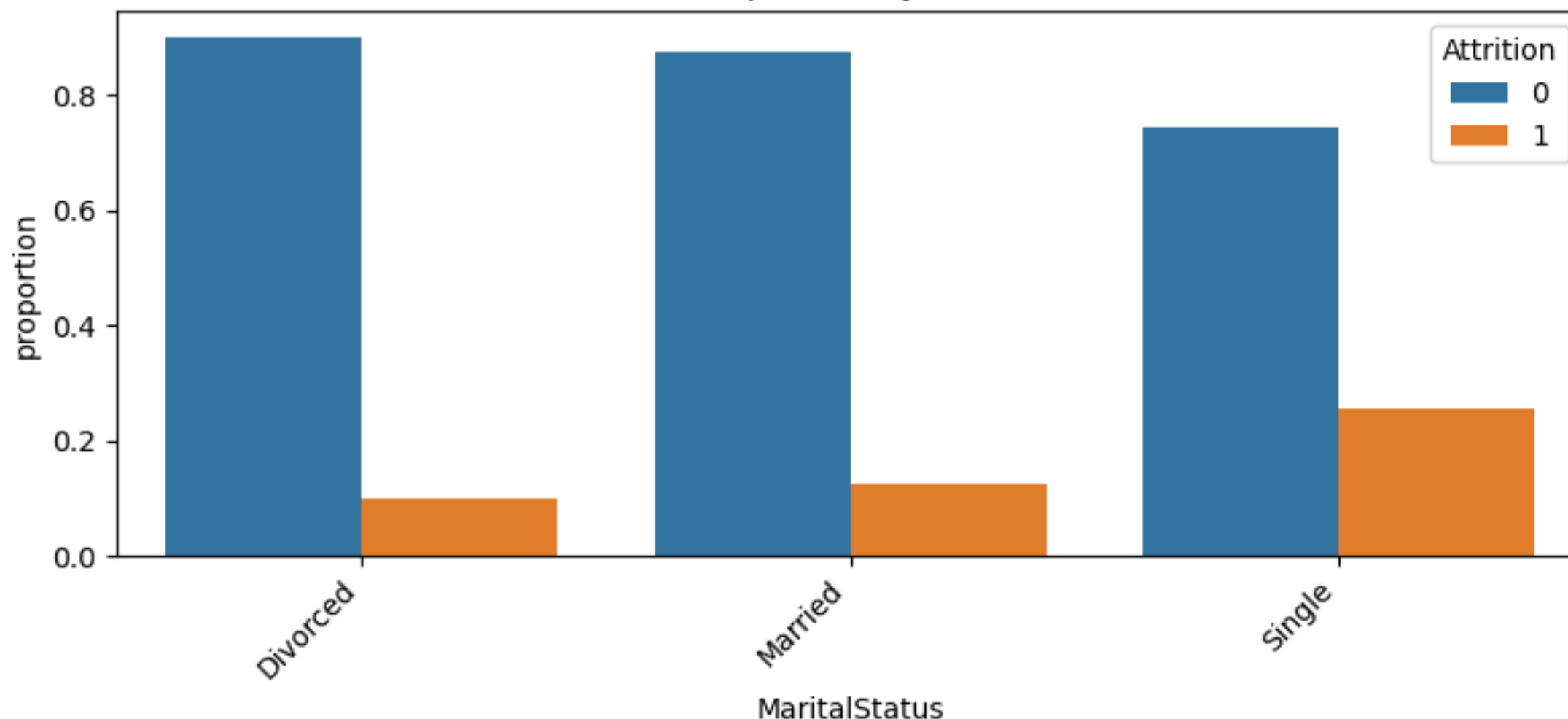
-MaritalStatus and OverTime



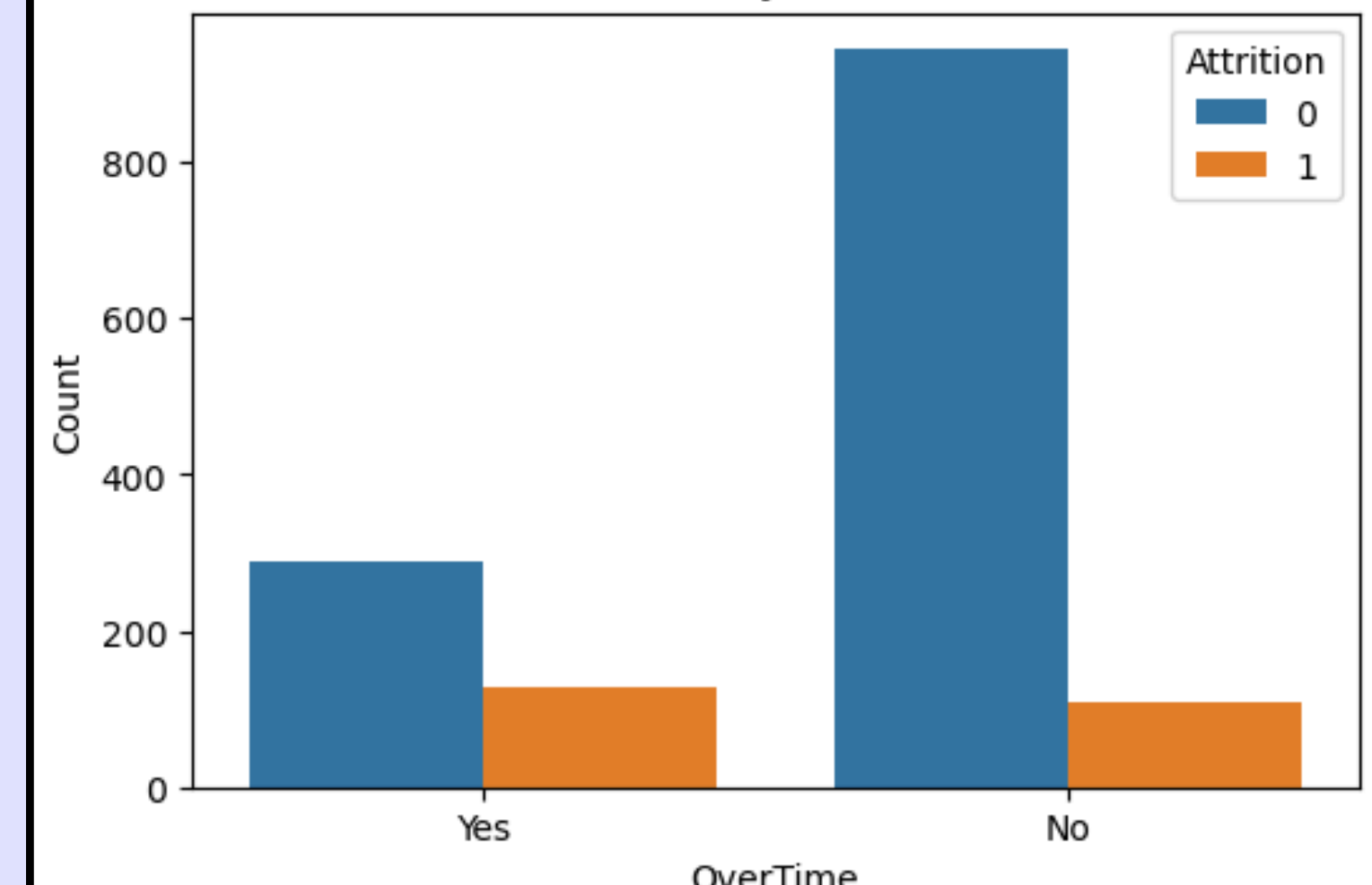
Insights and Analysis:

- Single employees have the highest attrition proportion.
 - Married and divorced employees show much lower attrition, suggesting greater stability.
 - Potentially life stage and personal responsibilities may influence retention.
 - Single employees may be more mobile or open to job changes.
- Employees who work overtime show noticeably higher attrition.
 - Non-overtime employees have much lower attrition, with most staying.
 - The gap between “Yes” and “No” Attrition categories suggests workload pressure is a major driver of turnover.
 - Overtime leads to potential burnout among overtime workers.

Attrition Proportion by MaritalStatus



Attrition by OverTime

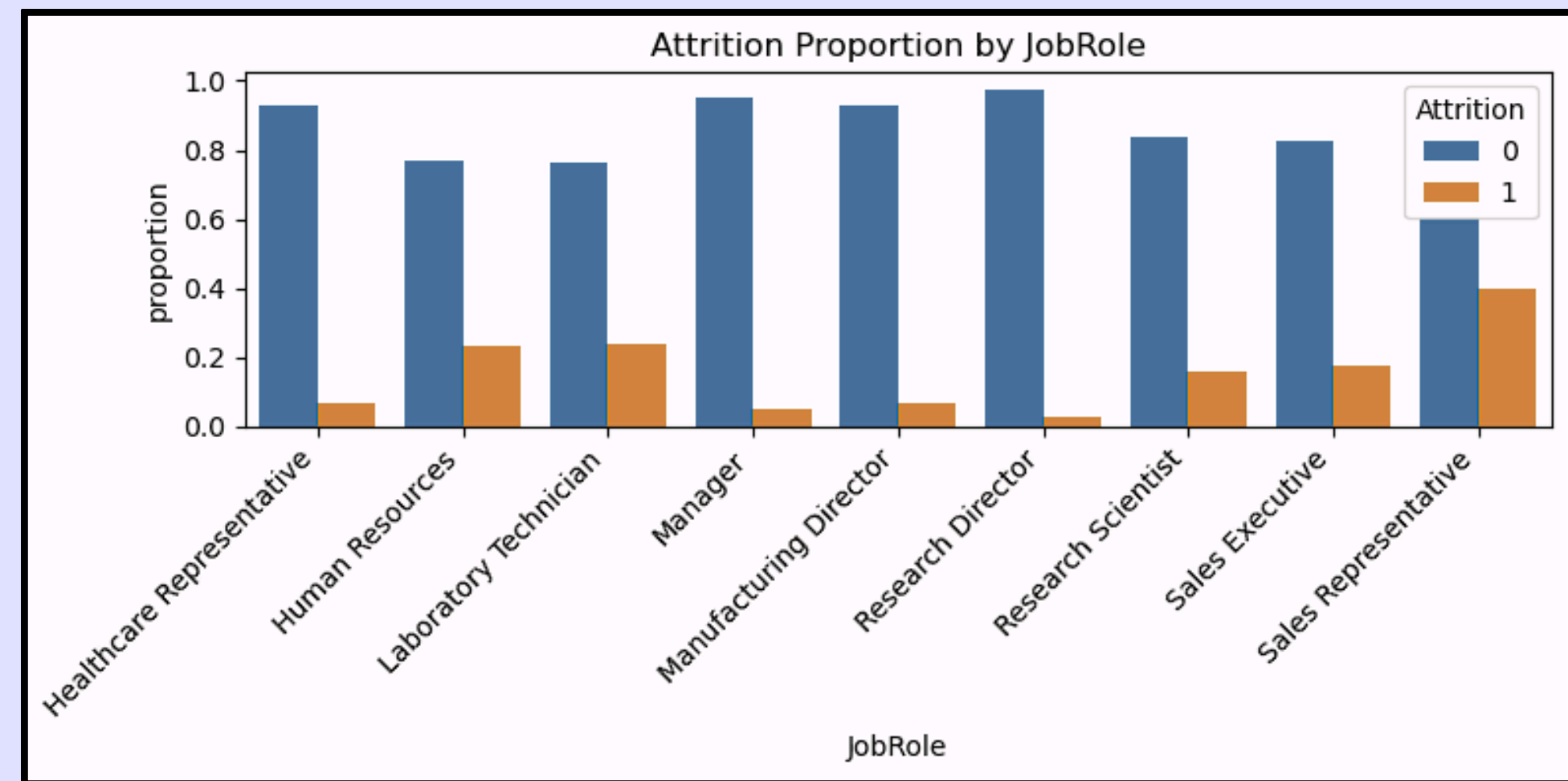
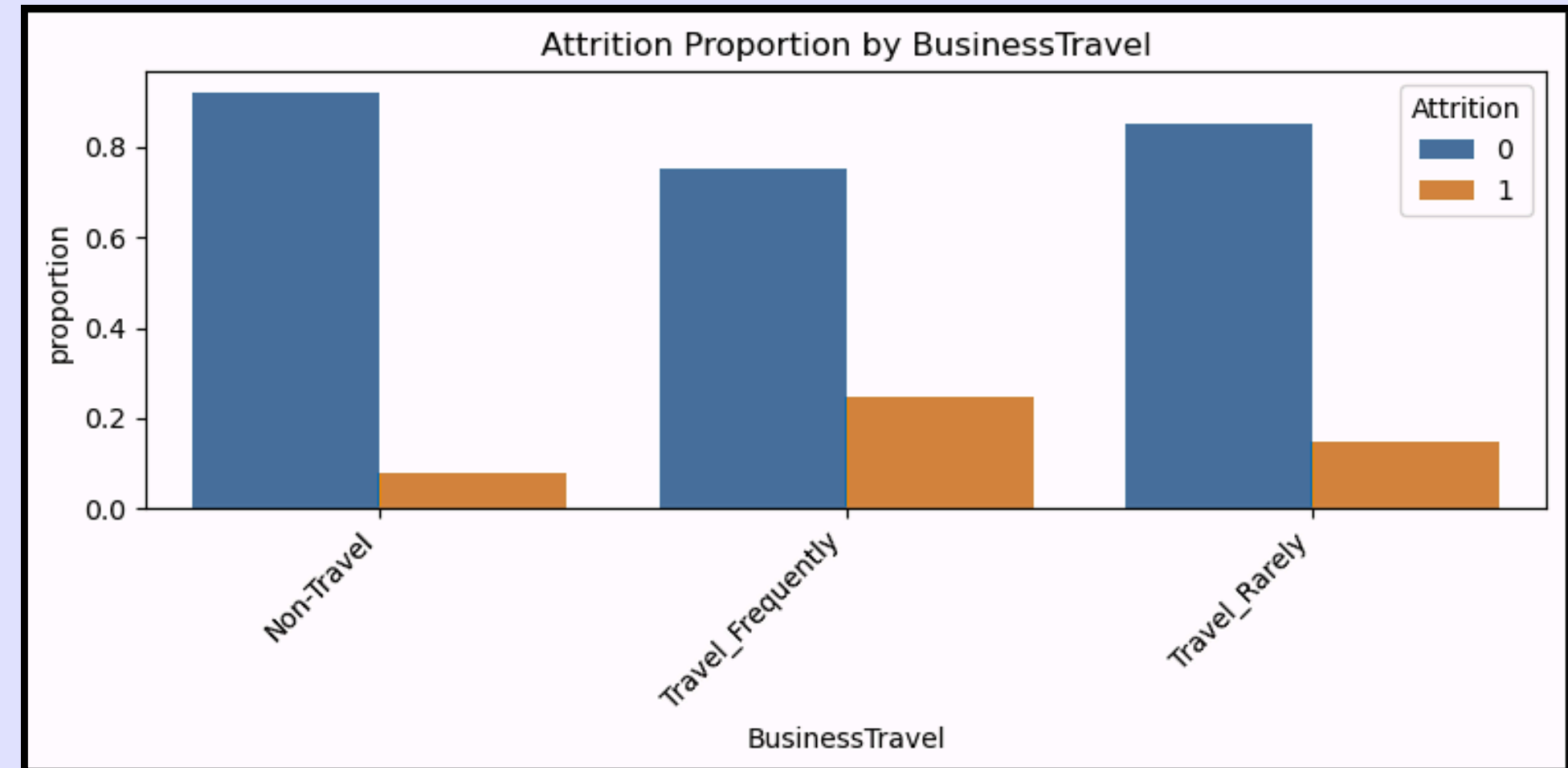


Bivariate Analysis

-BusinessTravel and JobRole

Insights and Analysis:

- Highest attrition among employees who travel frequently.
- Lowest attrition among non-travelers, indicating stability when travel demands are low.
- Pattern indicates that travel intensity is linked to burnout or job dissatisfaction.
- Sales Rep. and Human Resource roles show the highest attrition proportions.
- Technical and managerial roles like Research Director, Manager, Manufacturing Director shows much lower attrition.
- Indicates that customer-facing and administrative roles experience more turnover, possibly due to workload, stress, or job market mobility.
- Most other roles maintain high retention stability.

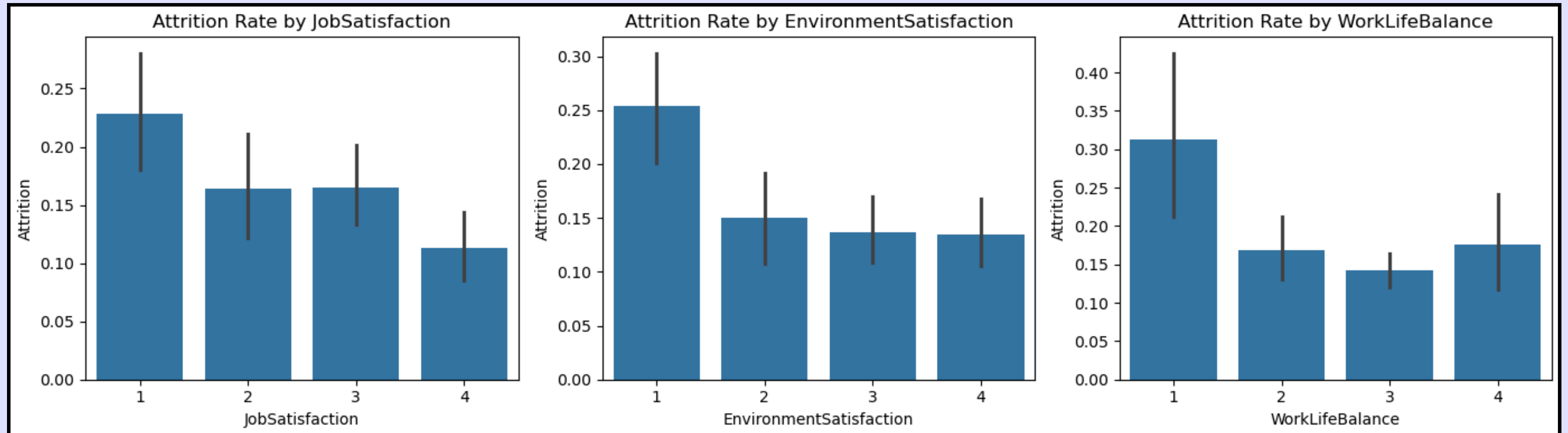
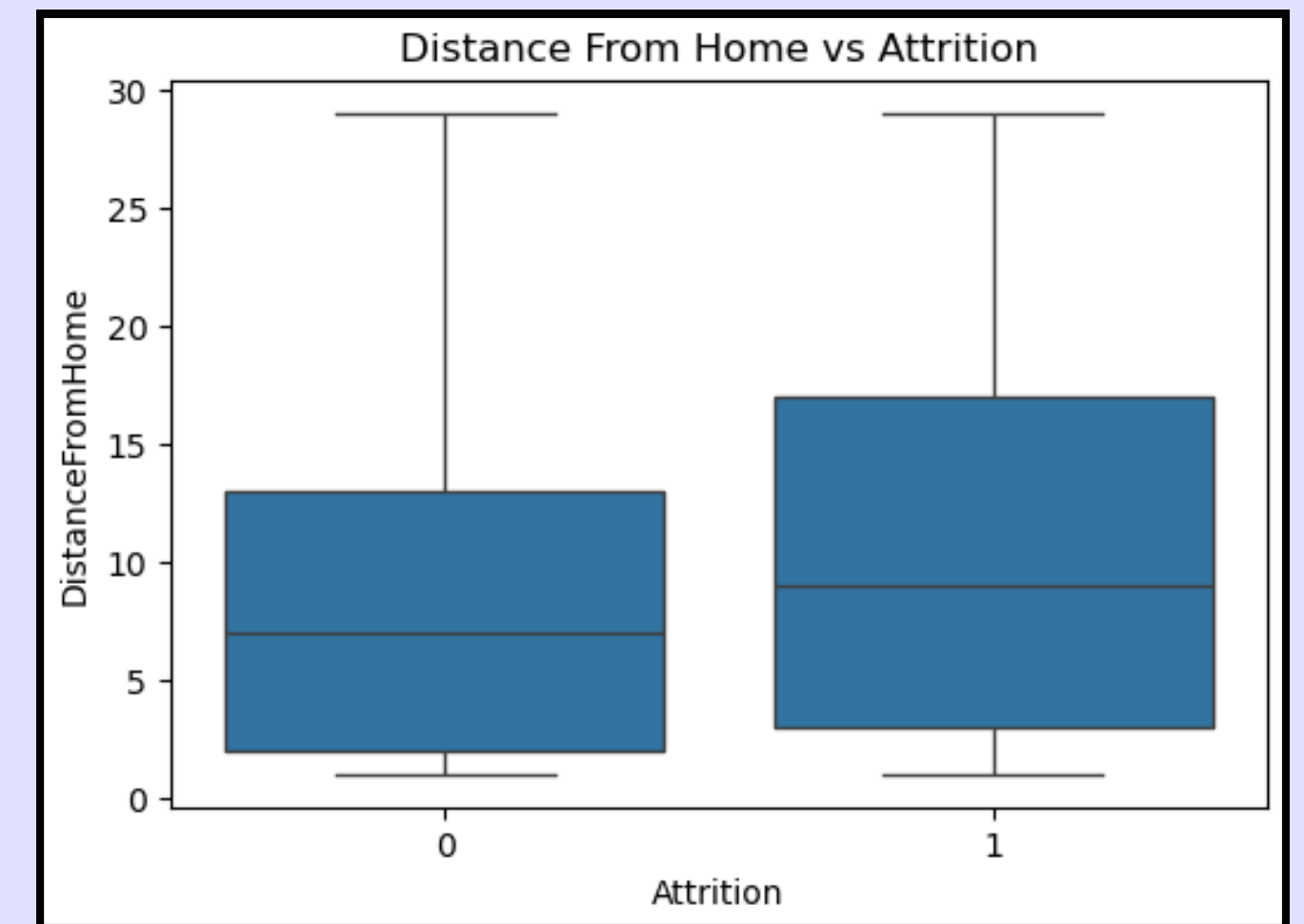


Bivariate Analysis

-JobSatisfaction, EnvironmentSatisfaction, DistanceFromHome, and WorkLifeBalance

Insights and Analysis:

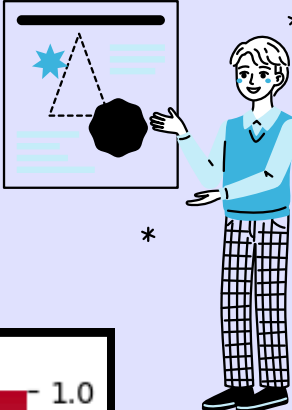
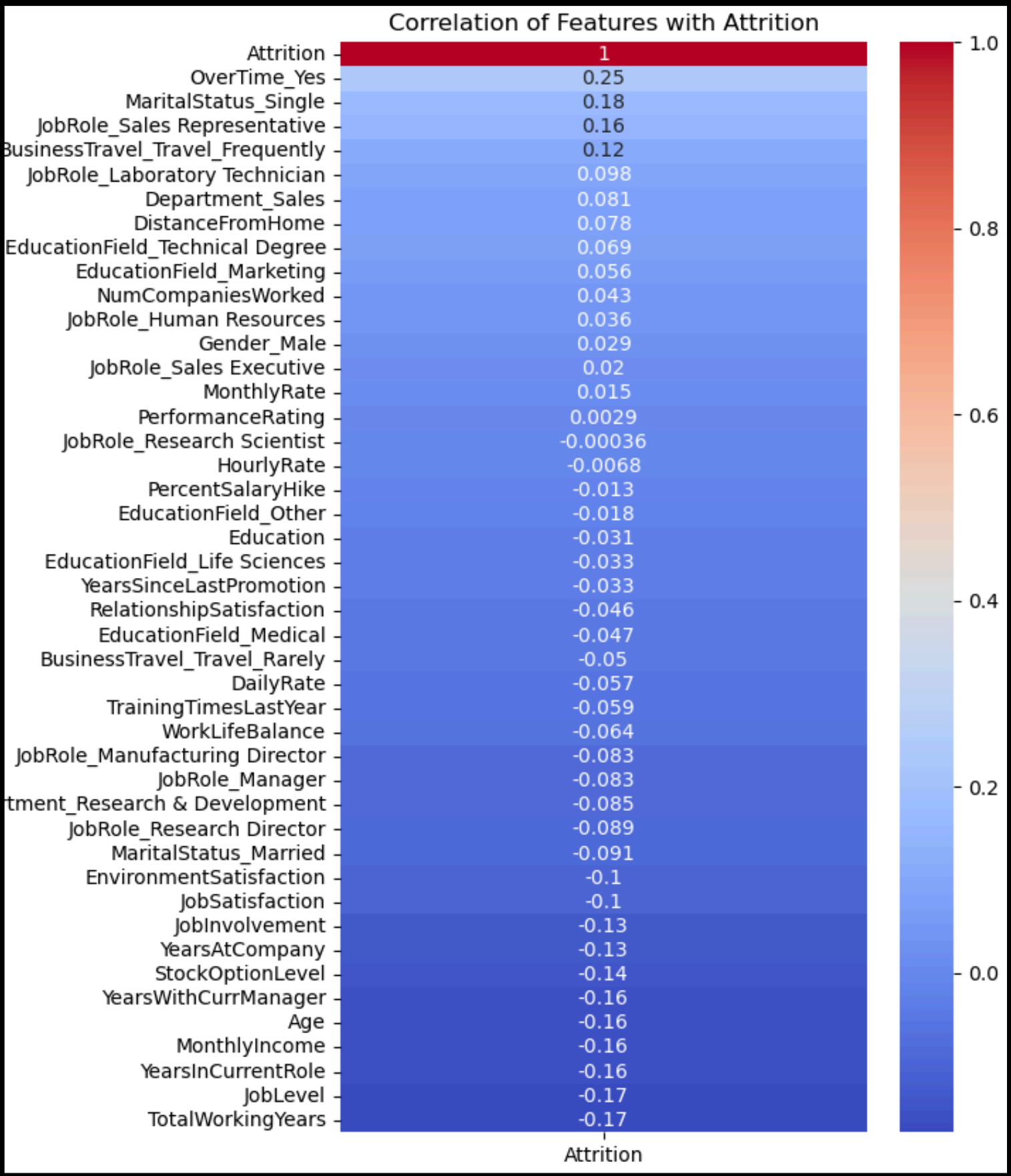
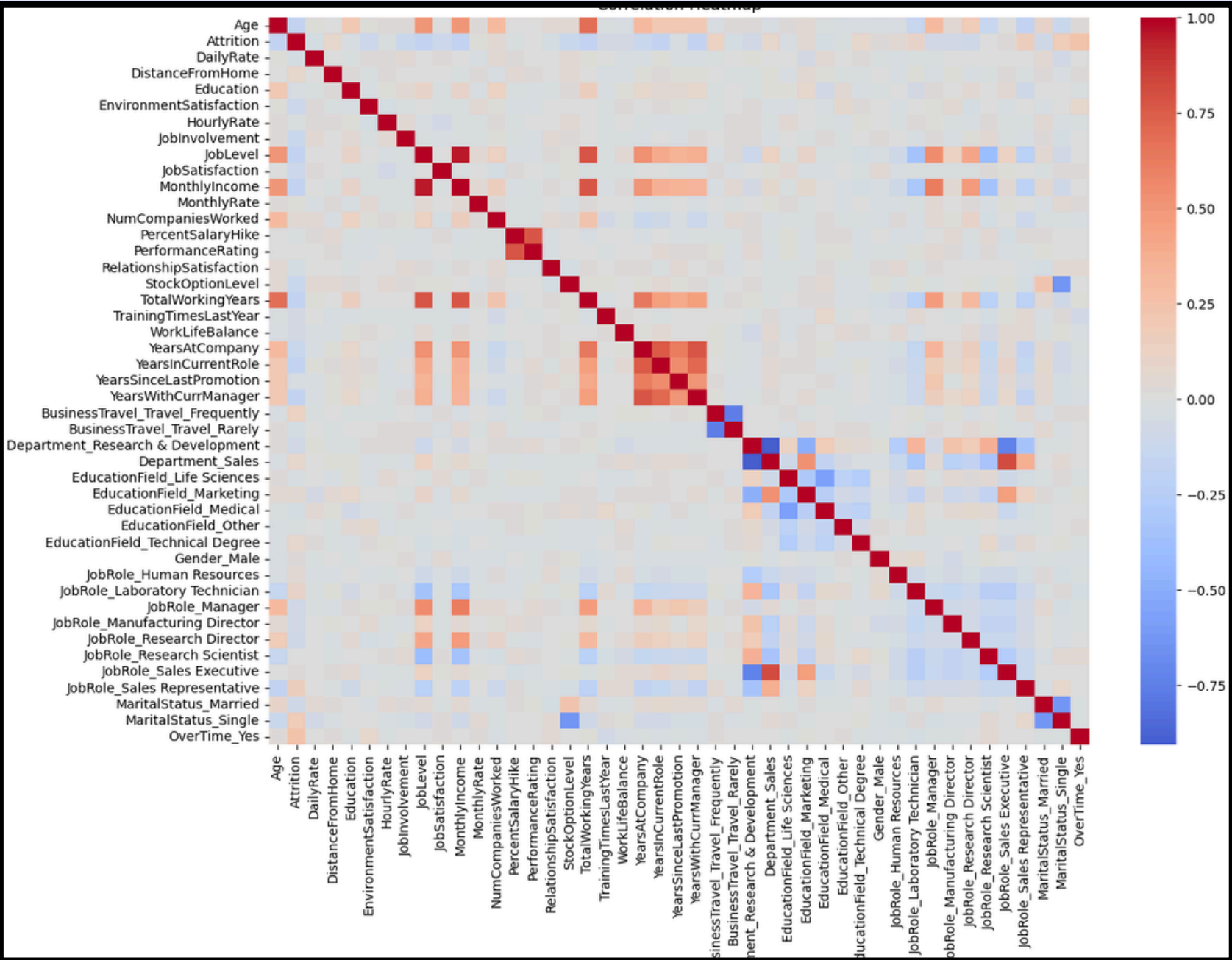
- Lower job satisfaction is strongly linked to higher attrition
- Poor work-life balance increases the likelihood of leaving
- Employees living farther from work show slightly higher attrition
- Satisfaction-related factors are key drivers of attrition
- Overall, overtime, job satisfaction, and work-life balance are the strongest factors influencing employee attrition.



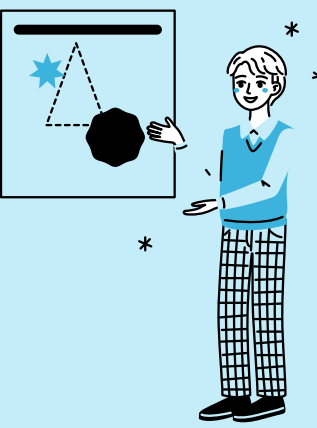
Multivariate Analysis

Insights and Analysis:

- The heatmap shows relationships between multiple features and attrition
- Overtime has a strong positive correlation with attrition
- Job satisfaction and environment satisfaction show negative correlation
- Monthly income and total working years are negatively related to attrition
- No single feature dominates — attrition is influenced by multiple interacting factors

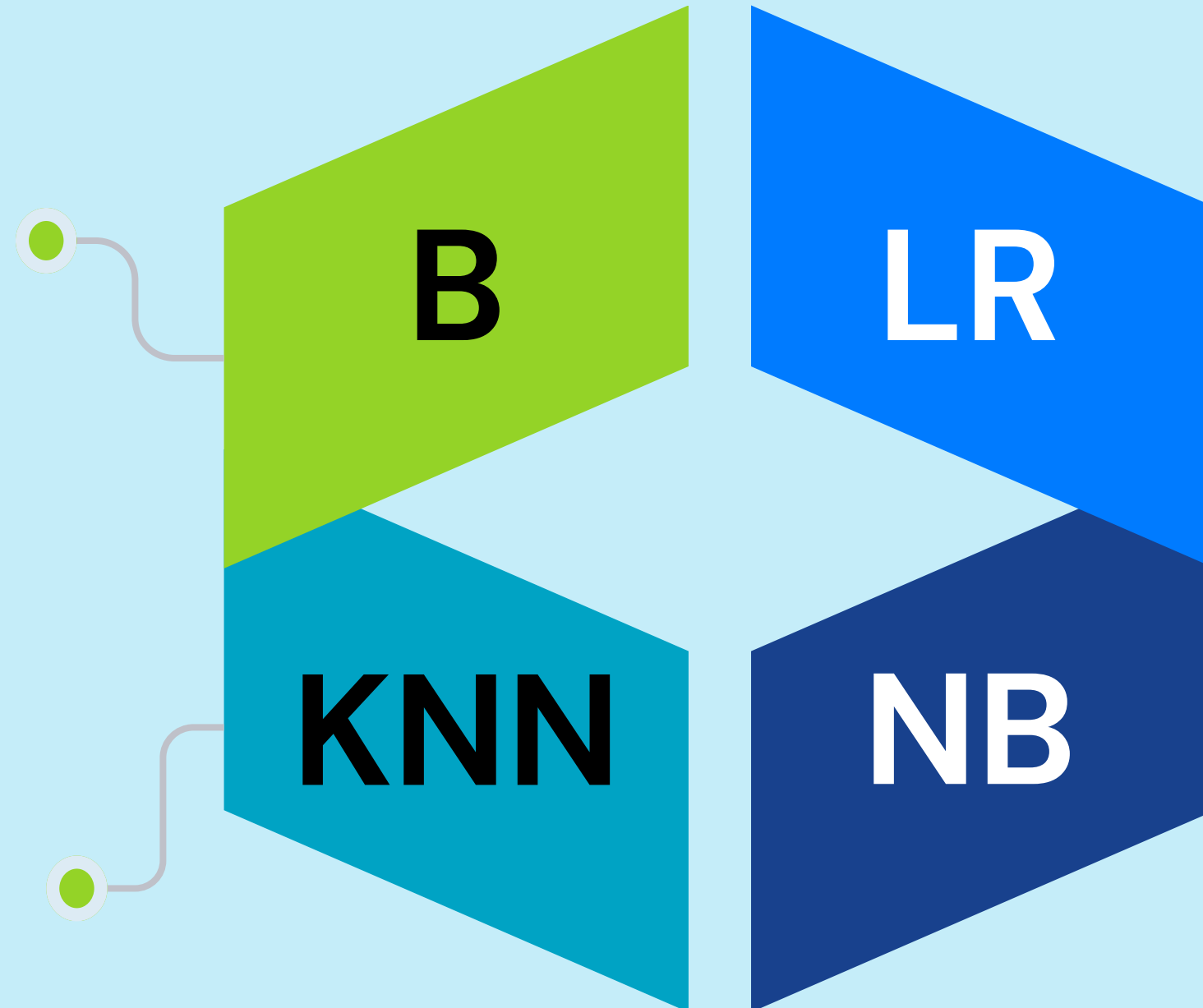


Model Selection



Baseline Model

- Benchmark model
- Majority-class predictor
- No use-case



Logistic Regression

- Probability based model
- Interpret features for classification
- Binary friendly

KNN Classification

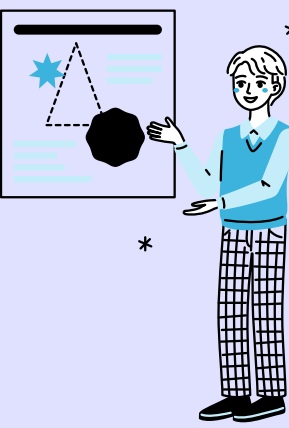
- Distance based model
- Capture non-linear patterns
- Simple logic

Naive Bayes Classification

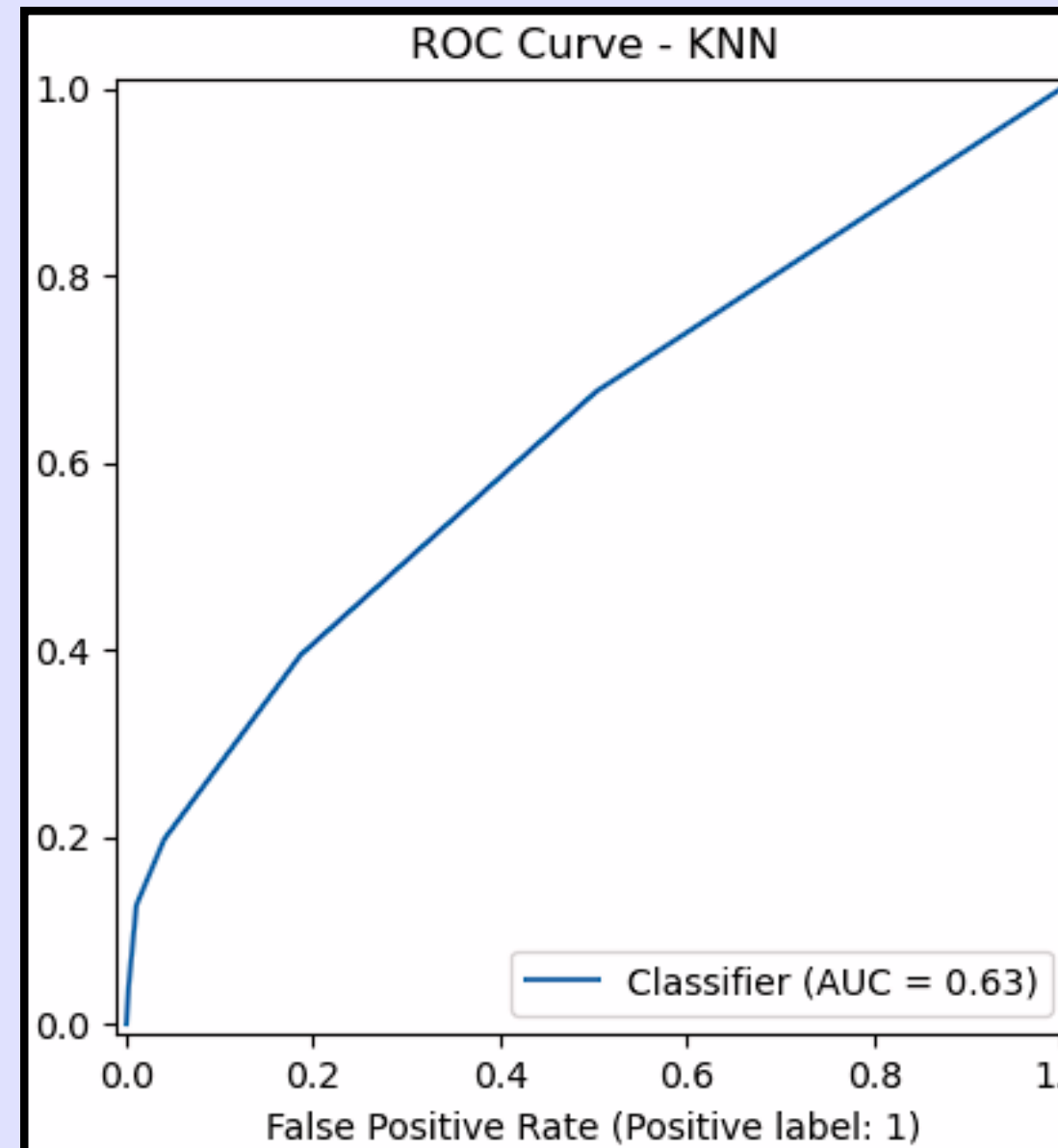
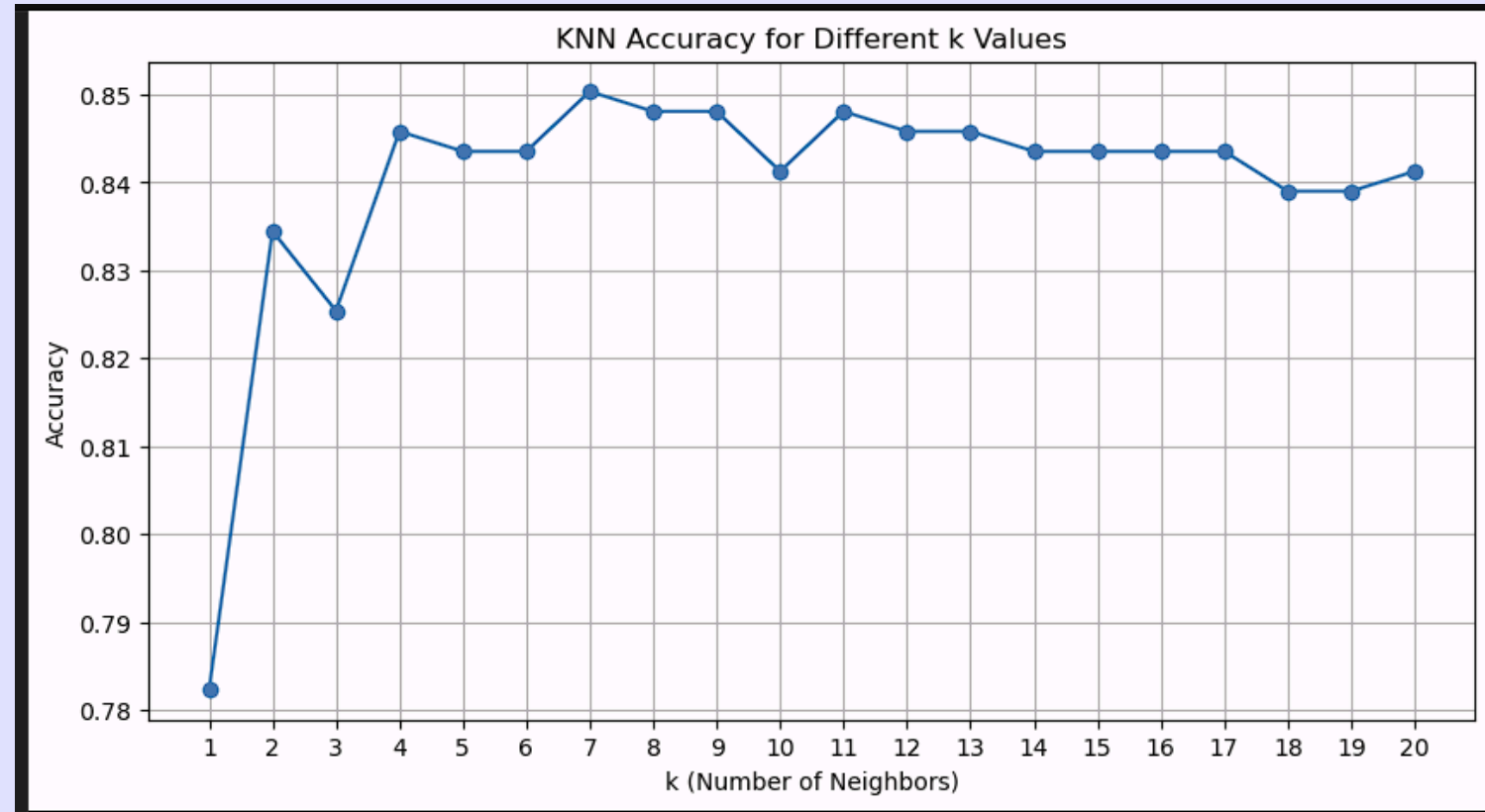
- Probabilistic classifier
- Independent features contribution
- Categorical variable friendly

Models Evaluation

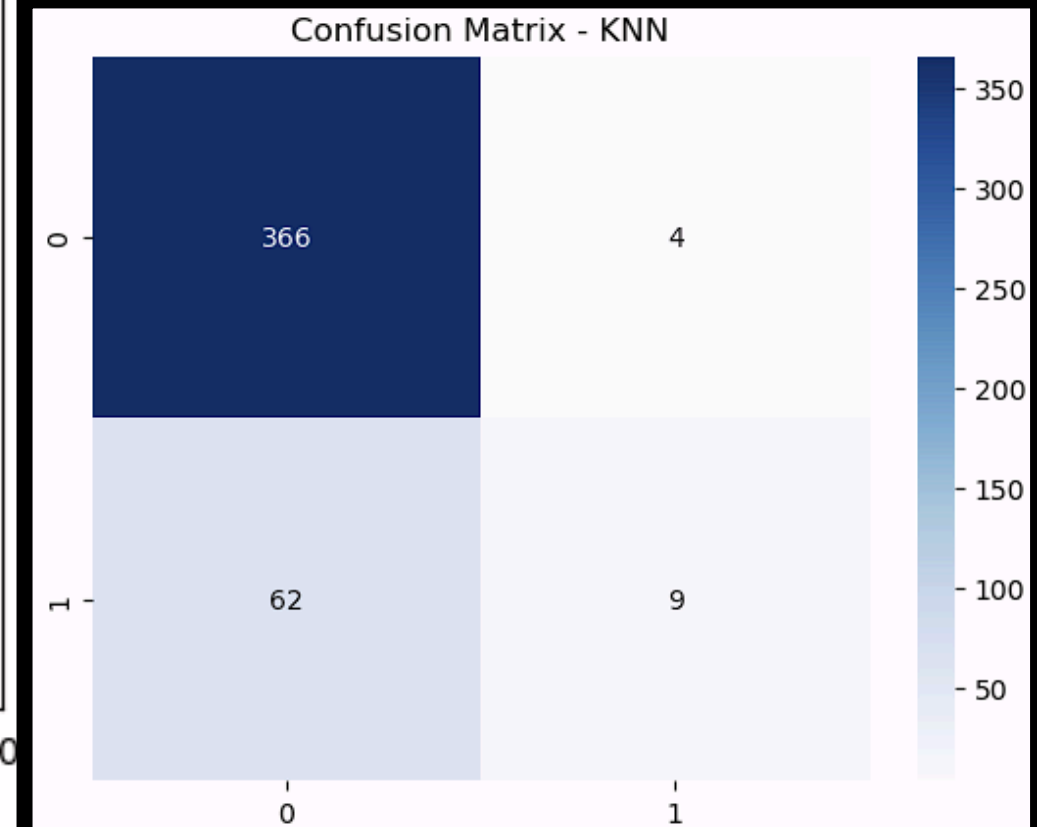
KNN Classification Model



Criteria for k value selection



=== KNN (k = 7) ===
Accuracy : 0.8503401360544217
Precision: 0.6923076923076923
Recall : 0.1267605633802817
F1 Score : 0.21428571428571427



Results

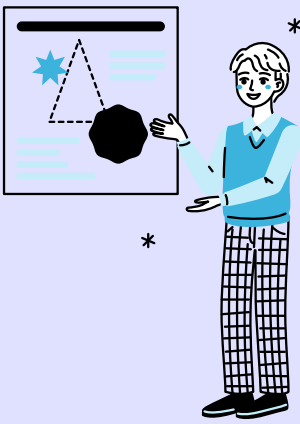
- Very High Accuracy
- Good Precision- Often correctly predicts
- Low Recall- misses actual attrition cases
- F1- poor balance between precision and recall
- High False Negative misses real attrition
- ROC shows moderate discrimination ability

Conclusion

Strong at predicting "No Attrition", weak at identifying leavers

Models Evaluation

Naive Bayes Classification

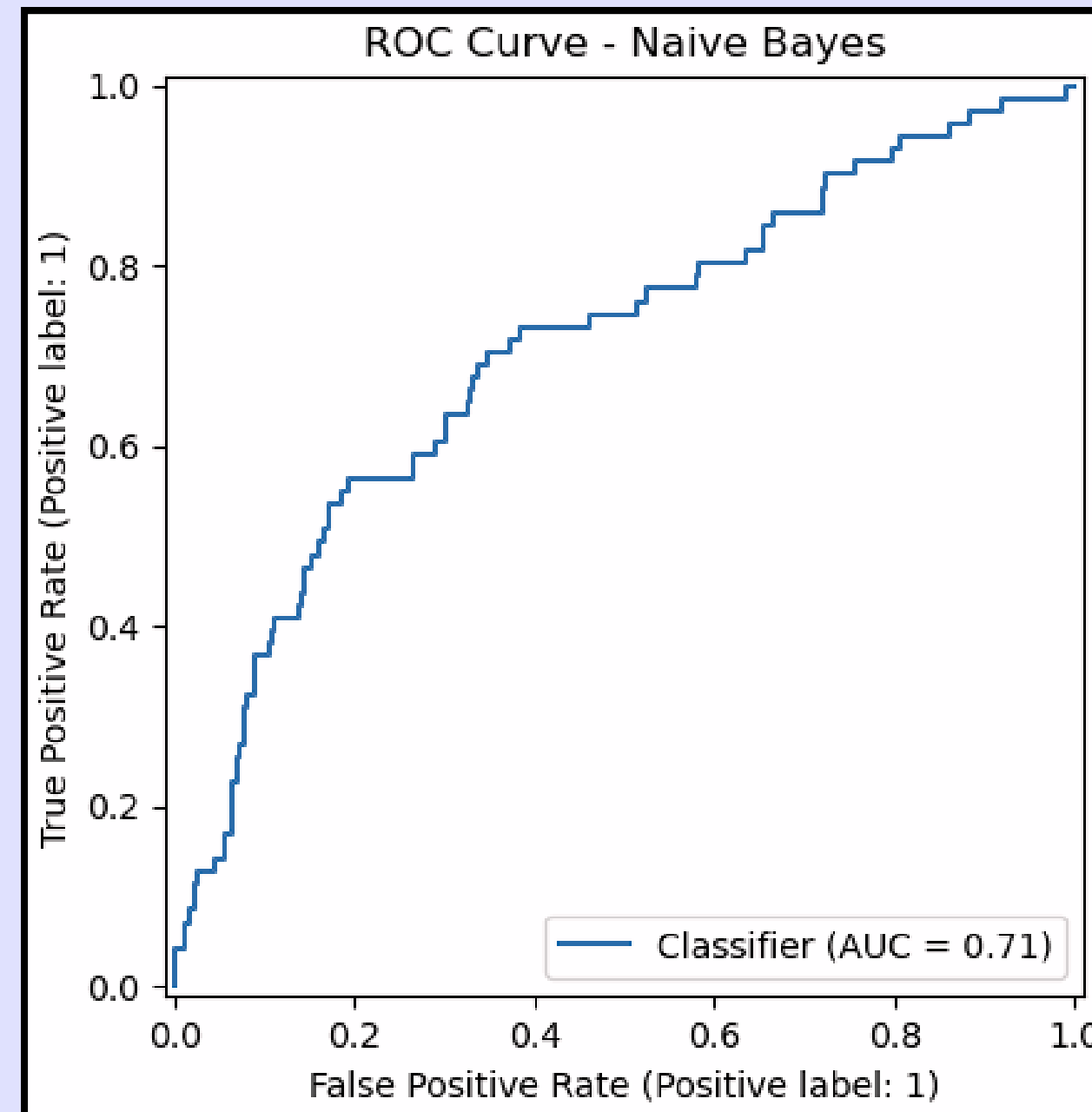


Results

- Moderate Accuracy
- Low Precision- unreliable in predicting attrition
- Strong Recall- good in identifying actual attrition cases
- High False Positive overpredicts attrition
- ROC shows decent discrimination ability

Conclusion

Model prioritizes predicting "Yes Attrition", even if it means more false alarms



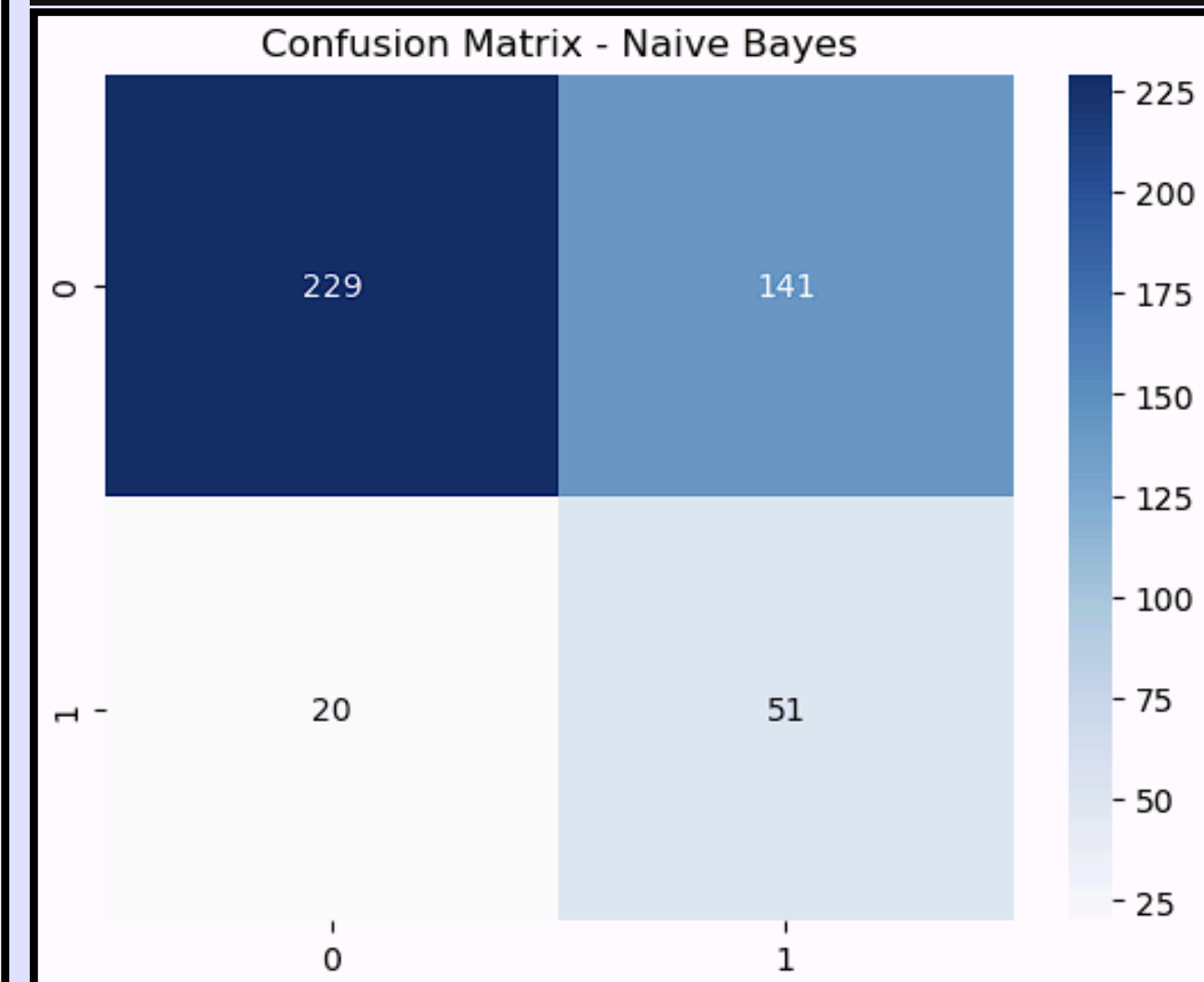
=== Naive Bayes ===

Accuracy : 0.6349206349206349

Precision: 0.265625

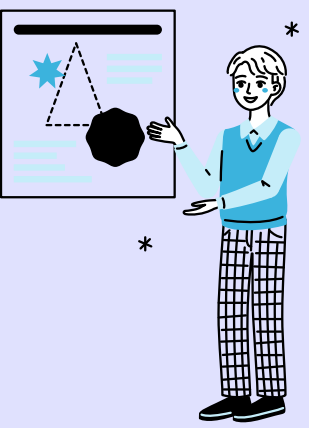
Recall : 0.7183098591549296

F1 Score : 0.38783269961977185



Models Evaluation

Logistic Regression

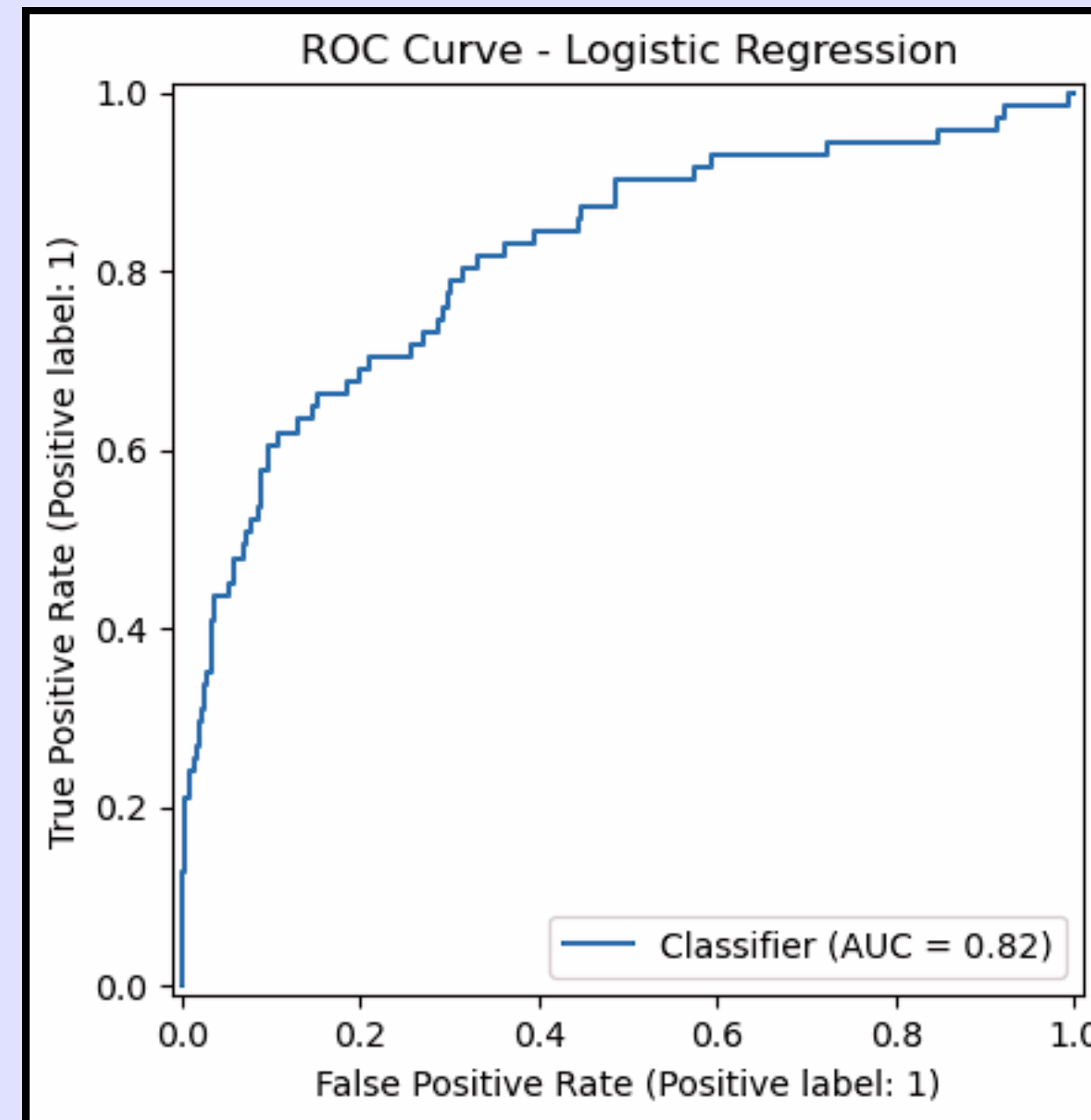


Results

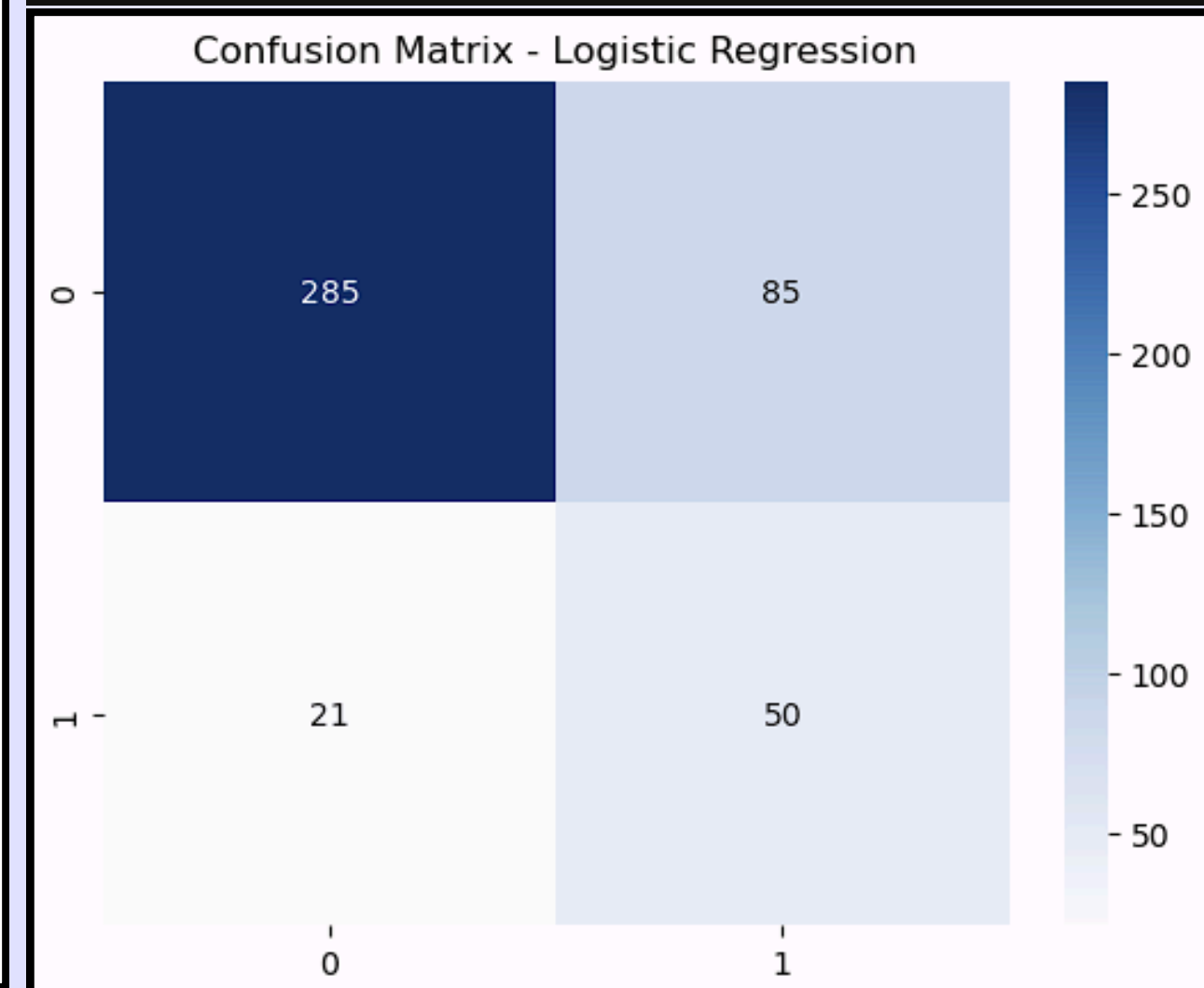
- High Accuracy
- Moderate Precision- satisfactory results in predicting attrition
- Strong Recall- good in identifying actual attrition cases
- Good balance between False Positive and False Negative
- ROC shows good discrimination ability

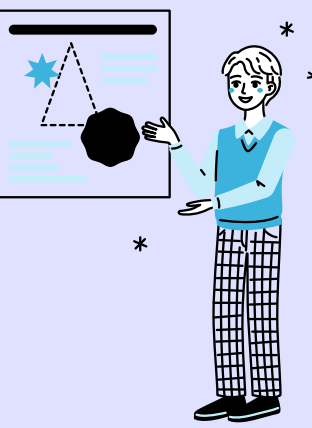
Conclusion

An overall balanced and reliable model in attrition prediction



=== Logistic Regression ===
Accuracy : 0.7596371882086168
Precision: 0.37037037037037035
Recall : 0.704225352112676
F1 Score : 0.4854368932038835





Models Evaluation

Evaluation Metric Model Table

	Model	Accuracy	Precision	Recall	F1 Score
0	Baseline	0.839	0.000	0.000	0.000
1	KNN	0.850	0.692	0.127	0.214
2	Naive Bayes	0.635	0.266	0.718	0.388
3	Logistic Regression	0.760	0.370	0.704	0.485

Decision

Logisitic Regression with the best overall performance of high accuracy, decent precision and high recall is our champion model

CONCLUSION

- Finding 1: Attrition is NOT random

Low income + overtime = more likely to leave

- Finding 2: Our model worked (kind of)

Better than guessing, but not perfect

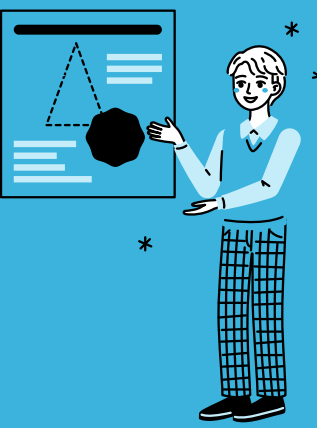
- Finding 3: ML can help with HR problems

You just need clean data + clear question

- Problem: Unbalanced data (more stayers than leavers)
- Model got good at predicting "stay" but not "leave"
- Main takeaway: Even a little warning is better than none
- HR can focus on keeping at-risk employees happy



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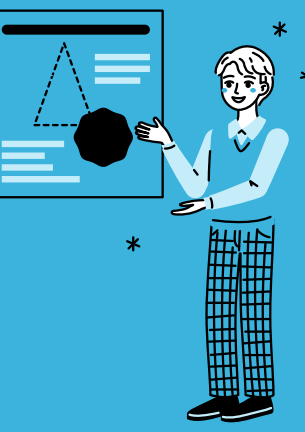
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THANK YOU